

1.		4
1.1.		4
1.2.		4
2.		5
2.1.		5
2.2.		5
2.3.		5
2.4.		6
2.5.		6
2.6.		6
2.7.		6
2.8.		7
3.		8
3.1.		8
3.2.		8
3.3.		8
3.4.		8
3.5.		9
3.6.		9
3.7.		9
3.8.		9
3.9.		10
3.10.		10
3.11.		10
3.12.		10
3.13.		11

3.14.		11
3.15.		11
3.16.		12
3.17.		12
3.18.		12
3.19.		13
3.20.		13
3.21.		14
3.22.		14
3.23. n		14
4.		15
4.1.		15
4.2.		15
4.3.		15
4.4.		15
4.5.		16
4.6.		16
4.7.		16
4.8.		17
4.9.		17
4.10.		17
4.11.		18
5.		19
5.1.		19
5.2.		19
5.3.		19
5.4.		19
5.5.		20
5.6.		20
5.7.		20
5.8.		21

6.		22
6.1.		22
6.2.		22
6.3.		22
6.4.		22
6.5.		22
6.6.		23
6.7.		24
6.8.		24
6.9.		24
6.10.		25
6.11.		25
6.12.		25
6.13.		26
6.14.		26
7.		27
7.1.		27
7.2.		27
7.3.		28
7.4.		28
7.5.		28
7.6.		28
7.7.		29
7.8.		29
7.9.		29
7.10.		29
7.11.		30
7.12.		30
7.13.	2	30
7.14.		31
7.15.		31

1.

1.1.

2022

pp. 40 - 41

1.2.

2022

pp. 43, 47 - 54

12 $X \text{ cm}^2$ $Y \text{ g}$

$$(1) \quad Y = aX + b$$

(2)

(3)

$(X, Y) = (11, 23), (23, 49), (5, 10), (50, 115), (32, 64), (21, 45)$
 $(16, 35), (39, 84), (25, 50), (45, 100), (5, 11), (34, 70)$

2.1.

(1)

2.2.

(4) 3

 $\mathcal{C}$

$$(4) \quad P(X \geq 4)$$

2.4.

2023

pp. 81, 86

$$f(x) = \begin{cases} \frac{c}{x^3} & (x \geq 10) \\ 0 & (x < 10) \end{cases}$$

- (1) c
- (2) $F(x)$
- (3) $P(x \geq a) = \frac{1}{2} \qquad a$

2.5.

2024

pp. 81, 86

c

$$f(x) = \begin{cases} c(1-x) & (0 \leq x \leq 1) \\ 0 & (\text{otherwise}) \end{cases}$$

- (1) c
- (2) $P\left(\frac{1}{2} < x \leq \frac{2}{3}\right)$
- (3) $F(x)$
- (4) $P\left(\frac{1}{2} \leq x < 3\right)$

2.6.

2022

2023

2024

pp. 134, 151, 180

- (1)
- (2)
- (3)
- (4)

2.7.

2023

2023

pp. 134, 151, 180

$X_1, X_2, \dots, X_n$

Y

2

- (1) $X_1, X_2, \dots, X_n$ Y
- (2) $2 \qquad /$
- (3)

- (4)
- (5)
- (6)1

2.8.

2025						
(1)	X	μ	σ^2			
(2)	200		60	6		42
	78					

3.

3.1.

2024				p. 107
6	4	1	2	
(1)	2			
(2)	2			

3.2.

2026				p. 107
	3	5		
(1)	1	1		
	(a)	2		
	(b)	2		
	(c)	2	1	
(2)			2	
	1			
(3)	(1)	(2)		

3.3.

2022				pp. 107 - 108
1	2	3	4	2
X_1	X_2	$Y = X_1 - X_2 $		
(1)	Y			

$$f(x, y) = \begin{cases} 6e^{-2x-3y} & (x \geq 0, y \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$$

$X \quad Y$

3.5.

2026

$X \quad Y$

$$\int_0^\infty e^{-x} dx = 1, \quad \int_0^\infty xe^{-x} dx = 1$$

$$(1) \quad f(x, y) = \begin{cases} xe^{-x-y} & (x \geq 0, y \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$$

$$(a) \quad f_X(x) \quad f_Y(y)$$

$$(b) \quad X \quad Y$$

$$(2) \quad f(x, y) = \begin{cases} c(x+y)e^{-x-y} & (x \geq 0, y \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$$

(1)

$1 \qquad \qquad \qquad 2$

(2)

$1 \qquad 2 \qquad \qquad \qquad A \qquad 2 \qquad 3$

$B \qquad \qquad \qquad A \qquad B$

(3)

$P(A|B)$

3.9.

2024

p. 115

$2 \qquad \qquad \qquad A \qquad \qquad \qquad 4 \qquad \qquad \qquad B \qquad \qquad \qquad 2$

$7 \qquad \qquad \qquad C \qquad \qquad \qquad 2 \qquad \qquad \qquad 9$

(1)

$A \qquad \qquad \qquad B$

(2)

$A \qquad \qquad \qquad C$

3.10.

2023

pp. 72 - 73, 109 - 110, 114 - 115

$2 \qquad \qquad \qquad E \qquad F \qquad \qquad \qquad P(E) = a \qquad P(F) = b$

(1)

$P(E \cup F) \qquad a \qquad b$

(2)

(1)	2	A	B						
(2)		500	1	X				X	
	2	K_1	K_2			3 : 1			
			K_1, K_2			0.98	0.90		
						0.01			
				K_1					

3.13.

2022	2023						pp. 118 - 119
	3	5		1			
	2						
(1)	1						
(2)	2		1				

3.14.

2023						pp. 118 - 119
	1	10%		3		
(1)						
(2)		2				

3.15.

2022						pp. 118 - 119
	5	4	3			
1		100 g	45 g	30 g		
(1)		1				
	40 g					

(2)	2	
	100 g	2
(3)	3	
	150 g	3

3.16.

2022	2024	pp. 118 - 119
	A	0.1%
A	/	
	99%	10%
(1)		
(2)	A	1

3.17.

2023		pp. 118 - 119
		99%
	99%	0.5%
	1	
(1)		
(2)		
(3)		
(4)		

3.18.

2023		pp. 118 - 119
	S	S
K		

•	S	K	99%	1%
•	S	K	99%	1%
(1)		K		S
(2)	(1)			

3.19.

2025

pp. 118 - 119

<i>A</i>	
<i>B</i>	
(1) $P(A B) = 0.98$ $P(A B^c) = 0.005$ $P(B) = 0.001$	
(2) $P(A B^c) = 0.003$ $P(B) = 0.0005$ $P(B A) \geq 0.142$	$P(A B)$

3.20.

pp. 118 - 119

500	1	XXX	2
K_1 K_2		$K_1 : K_2 = 3 : 1$	
K_1		0.98	
K_2		0.90	
		0.01	
(1)		K_1	
(2) (1)			

3.21.

2023

pp. 122 - 124

p

$0 < p < 1$

X

Y

$f(k) = \begin{cases} p & (k = 1) \\ 1 - p & (k = -1) \end{cases}$

$Z = XY$

(1)

$E[Z]$

(2)

X

Z

$E[(X - E[X])(Z - E[Z])]$

(3)

X

Z

p

p

Y

Z

(4)

(3)

p

$\Pr[X + Y + Z \leq 2]$

3.22.

2023

2024

pp. 125 - 126

2

X

Y

(1)

a

(2)

X

Y

$\text{Cov}(X, Y)$

(3)

X

Y

	$Y = 1$	$Y = 10$	$Y = 100$
$X = 1$	a	a	a
$X = 2$	0	a	a
$X = 3$	a	a	a

3.23. n

2023

p. 127

n

1

n

$(n + 1)$

4.

4.1.

2025

2

$B(2,p)$

4.2.

2024

p. 141

1

6

(1)

10

5

(2)

100

25

4.3.

2023

2026

pp. 93 - 95, 141

p

n

X

2

$\text{Bin}(n,p)$

$q = \left(\frac{X}{n}\right)$

(1)

$P(|q - p| \leq 0.02) \geq 0.95$

n

(2)

(1)

n

(2)

(3)

p

(3)

(1)

-

n

(4)

(2)

(3)

n

4.4.

2024

pp. 143 - 145, 153

(1)

:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (-\infty < x < \infty)$$

$y = f(x)$

(x,y)

$$(2) \quad \quad \quad /$$

$$(3)$$

4.5.

2022

pp. 121 - 122, 143

$$(1) \quad X \quad Y \quad \quad \quad N(170, 6^2)$$

$$(a) \quad W = \frac{X + Y}{2}$$

$$(b) \quad \quad X \quad Y \quad \quad \quad W$$

$$(2) \quad X \quad Y$$

$$f(x) = \frac{1}{2\pi} \quad (0 \leq x \leq 2\pi), \quad g(x) = \frac{1}{12} \quad (-3 \leq x \leq 9)$$

$$W = XY$$

4.6.

2024

pp. 147 - 151

$$(1) \quad X \quad \quad \quad N(\mu, \sigma^2)$$

$$(a) \quad P(\mu - \sigma \leq X \leq \mu + \sigma)$$

$$(b) \quad P(\mu - 2\sigma \leq X \leq \mu + 2\sigma)$$

$$(c) \quad P(\mu - 3\sigma \leq X \leq \mu + 3\sigma)$$

$$(2) \quad \quad \quad ($$

$$)$$

$$(3)$$

$$(a) \quad z(0.01)$$

$$(b) \quad \Phi(z_0) = 0.01 \quad \quad \quad z_0$$

$$(c) \quad z(0.005)$$

4.7.

2024

pp. 149 - 151

$$(1) \quad a \quad \quad \quad Z \quad \quad \quad N(0, 1)$$

$$P(|Z| \geq a) = 2P(Z \geq a)$$

(2)

X

$N(m, \sigma^2)$

$P\left(|X - m| \geq \frac{\sigma}{4}\right)$

4

(3)

m

σ

n

$\overline{X}$

$P\left(|\overline{X} - m| \geq \frac{\sigma}{4}\right) \leq 0.02$

n

4.8.

2024

2026

pp. 149 - 150

4

E

350,198

66.31

23.55

X

$N(66.31, (23.55)^2)$

(1)

$P(X \geq 80)$

(2)

(3-a)

(3)

80

95

(4)

20%

4.9.

2025

pp. 149 - 150

4

120

10

100

15

C

(1)

95%

C

(2)

4.10.

2022

pp. 153 - 154

- (1) 6
- (2) 10000
- (3) 1000049005100

4.11.

pp. 93 - 95, 153 - 154

200606

X

(1)

$42 \leq X \leq 78$

(2)

(3)

X

$N(60, 6^2)$

$P(42 \leq X \leq 78)$

5.

5.1.

20232025

pp. 168 - 169

2

5.2.

2024

pp. 93, 168 - 169

1ATMATM140

ATM160

ATM

/

60

75

90

105

120

5.3.

2023

pp. 166 - 169

(1)35

(2)1214

5.4.

2025

4

(1)13

2

(2)

X

$P(X \leq 3)$

$P(X = 3) = 5P(X = 5)$

5.5.

2023

pp. 166 - 169

1%

200

(1)

X

(2)

(3)

3

(2)

(3)

5.6.

2024

pp. 153 - 154, 166 - 169

4

(1)

240

6

38

45

(2)

0.1%

500

3

5.7.

2026

$p = 0.001$

$n = 500$

X

$e^{-0.5} \approx 0.6065,$

$\sqrt{0.5} \approx 0.7071$

(1)

(a)

X

(b)

$E[X]$

$\text{Var}(X)$

(2)

(a)

(b)

$P(X \geq 3)$

(3)

(a)

(b)

$$P(X \geq 3)$$

(4)

(a)

(2)

(3)

(b)

$$np$$

5.8.

2022

2023

pp. 139 - 143, 168 - 169

p

$0 < p < 1$

$(1 - p)$

N

$N > 1$

(1) $N = 5$

(2) $t \quad 0 \leq t \leq N$

(3) $m \quad m =$

$N \quad 0 \leq m \leq N \quad m$

(4) $n \quad 0 \leq n \leq N \quad N - n \quad P(n\&N, p)$

(5) (4) n

(6) $n \quad \lambda = Np \quad N \rightarrow \infty \quad (4) \quad P(n\&N, p)$

$$P(n; \lambda) \quad \lim_{N \rightarrow \infty} \left(1 + \frac{x}{N}\right)^N = e^x$$

6.

6.1.

2022

p. 174

2

(1)

(2)

6.2.

2025

μ

σ^2

n

$X_1, X_2, ..., X_n$

.

(1) $X_1, X_2, ..., X_n$

(2) .

(3) n $S_n = X_1 + X_2 + \cdots + X_n$

.

(4)

6.3.

2025

p. 179

1180

20

(1) 25 1170

(2) N 0.9 1175

N

6.4.

2025

1.2 kg

0.1 kg

100

122.5 kg

6.5.

2022

pp. 202 - 205

μ

σ^2

3

$X_1 \quad X_2 \quad X_3$

(1) .

(a) $T_1 = X_1 + X_2 - X_3$

(b) $T_2 = \frac{X_1 + 2X_2 + X_3}{4}$

(c) $T_3 = \frac{X_1 + X_2 + X_3}{3}$

μ

(2) $\{T_1, T_2, T_3\}$

6.6.

2023

2023

2025

pp. 176 - 180, 202 - 205

(1)

0

1.2

3

$\{-1, 0, 1\}$

(a)

(b)

(c)

(2)

(3)

(4)

θ

$\Theta_n = T(X_1, X_2, ..., X_n)$

3

(a)

(b)

(c)

(5)

3

6.7.

20222024

p. 205

100010

32, 86, 78, 49, 48, 77, 97, 49, 51, 74

1000

6.8.

2025

pp. 179, 216

p

n

X

(1) X .

(2) n X

(3) p_0 95%

n

(4) $p = 0.05$ $n = 1900$ $P(76 \leq X \leq$
114)

6.9.

2022

pp. 180 - 182

p

p 10,000

3,500

$p = \frac{3500}{10000} = 0.35$

(1) p

(2) p 2

6.10.

2024	2024	2025	pp. 224 - 225		
(1)	$Bin(1, p)$		$X_1, X_2, \dots, X_n$		
	(a)				
	(b)				
(2)	n	A	Y		
	(a)	$q = \frac{Y}{n}$	A	p	$100(1 - \alpha)\%$
	(b)				
(3)	S	TV	T		
	140	32	T		
	(a)		p	95%	
	(b)	p		3%	0.95
	(c)				

6.11.

2024	pp. 223 - 225				
(1)	p	0.8	0.9		0.99
	$\hat{p}$		p	0.02	
(2)	p			0.95	$\hat{p}$
	p		0.03		

6.12.

2025					
2	p	S	$1 - p$	F	
m			S	k	$B(m; p)$

- (1)

$B(m;p)$

$f(k)$

.
- (2)

X

$B(m;p)$

X

.
- (3)

p

$l(p)$

.
- (4)

p

.
- (5)

.

.
- (a)

(4)

?
- (b)

(4)

?
- (c)

~~(4)~~

—————~~?~~ ←

6.13.

- 2025
- $N(\mu,\sigma^2)$
- μ
- σ^2
- (1)

μ
- (2)

σ^2

6.14.

- 2025
- (1)
- (2)

/

σ^2

n
- N
- (3)

/

7.

- ① 仮説の検定
- ② 統計量の検定
- ③ 優位水準と棄却域の設定
- ④ 帰無仮説の棄却・選択

7.1.

2022

2023

pp. 227 - 231

(1)

(2)

(3)

(4)

5%

(3)

5%

(5)

(1)

(1)

(6)

(6)

(7)

• (8:)

• (9:)

7.2.

2022

pp. 227 - 234

(1)

(2)

7.3.

2023					pp. 2235 - 236
		8000			
	25			7700	
			5%		
		640			

7.4.

2023					pp. 238 - 240
	T				
S		365			6
		330, 350, 385, 321, 340, 365			
	5%				

7.5.

2025					pp. 238 - 240
	T				
S		360			6
		350, 355, 385, 390, 370, 365			

- (1)10%
- (2)5%

7.6.

2023	2024	2024	2025		pp. 235 - 236
(1)	μ		$N(\mu, 49)$	49	
	$H_0 : \mu = 4,$	$H_1 : \mu > 4$			
	m				
	(a)	0.05	$m = 6$	2	

(b)0.01 $m = 7$ 2

(2) $H_1 : \mu = 5$ 0.05

20.20.1

7.7.

2022pp. 241 - 242

(1) 2A B 2

10 12

A2.040.03B

2.020.022

5%

2

(2) A B

5%

7.8.

2022pp. 241 - 242

10070

6815

5%

7.9.

2024pp. 252 - 253

3 1

(1) A 10A

3 1A

5%

(2) 10 73 1

7.10.

2024pp. 252 - 253

5

1

5%

7.11.

2023

pp. 252 - 253

10

X

$H_0 : p = \frac{1}{2}$

H_0

p

(1) X

(2) X

(a) $H_1 : p < \frac{1}{2}$

5%

(b) $H_1 : p \neq \frac{1}{2}$

2%

7.12.

2023

pp. 252 - 253

5

1

p

(1) 5

5

0

(a) $p = \frac{1}{2}$

(b) $p = \frac{2}{3}$

(2) $p \geq \frac{1}{2}$

(a) 3

$p = \frac{1}{2}$

(b) $p = \frac{3}{4}$

3

7.13. 2

256

116

5%

(1) 2

.

(2)

2

10%

7.14.

2023

2025

pp. 252 - 253

$H_0 : p = \frac{1}{2}$

3

p

H_0

$0 < p < 1$

$\frac{1}{3}$

$\frac{2}{3}$

H_0

(1)

1

1

(2)

2

2

(3)

2

$\frac{1}{3}$

p

7.15.

2024

pp. 252 - 253

3

1

A

B

C

A

3

B

2

C

1

1

(1)

A

A

$\frac{1}{16}$

B

B

$\frac{1}{9}$

C

C

$\frac{1}{25}$

1

(2)

960

640

(1)

5%